

Optimal Correction of Environmental Stochastic Externalities

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Abstract

Pollution raises climate risk by shifting the distribution of everyday climate shocks, making extreme weather events more frequent and more intense, and by increasing the probability of an irreversible climate collapse. I study the impact of the first channel on the social cost of carbon, introducing what I call stochastic externalities, in which pollution affects the distribution of random climate damages. Solving the social planner's problem, first in a three-period model and then extending the analysis to an infinite-horizon recursive model, I show that the social cost of carbon gains an additional correction term, equal to the expected marginal damages weighted by the distribution's sensitivity to pollution. This term has closed forms for standard distribution families and nests the standard social cost of carbon. In a DICE-SE model calibrated to 1980-2024 disaster losses, I show that this new channel raises the social cost of carbon by 6.2% in a conservative calibration.

Keywords: *Optimal environmental taxation; stochastic externalities; carbon tax; endogenous climate risk; social cost of carbon; Pigouvian taxation*

JEL Classification: *D62; D82; H21; H23*

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1 Introduction

Pollution raises climate risk by shifting the distribution of everyday climate shocks, making extreme weather events more frequent and more intense, and by increasing the probability of an irreversible climate collapse. The latter channel has been extensively studied in climate economics, but the former has received less attention. In this paper, I study the impact of the first channel on the social cost of carbon, introducing what I call stochastic externalities, in which pollution affects the distribution of random climate damages. The costs of this channel are already large and growing. In the United States, the number of billion-dollar extreme weather events averaged 8.5 per year over 1980-2023, but grew to an average of 20.4 per year over 2019-2023 (Smith, 2020). Using NOAA simulation data, Bakkensen and Barrage (2026) moreover finds that climate change will roughly triple welfare losses from cyclones in the United States and the Caribbean.

In this paper, I develop a framework to study how an endogenous shift in the distribution of climate shocks affects the social cost of carbon. I model what I call stochastic externalities, where pollution affects the distribution of random climate damages, capturing the idea that pollution makes events such as floods, droughts, and wildfires more likely and more intense. The concept of weather events being represented as draws from a distribution shifting with pollution is common in Earth sciences (Naveau et al., 2020; Stott et al., 2016), but less so in economics. I close this gap and provide implications of this mechanism on the social cost of carbon. To build intuition, I first build a general three-period model where current emissions affect resources. The social cost of carbon is then the sum of a standard Pigouvian term and of a stochastic externality correction term. This correction term can be written in different equivalent ways. One is the score form, representing how the probability of each shock realization changes with pollution. Another is the quantile form, representing how much worse or better each quantile of the shock distribution becomes after the shift.

I then show three examples translating this term into economically meaningful channels. Under normally distributed shocks, the correction splits into mean-shift, risk-aversion, and damage-curvature channels. Under Pareto shocks, there is a single tail-fattening channel. This term can grow without bound faster than the standard Pigouvian term, reinforcing Weitzman's Dismal Theorem concerns. Under a regime-mixture distribution echoing the "tipping point" literature, the correction term is the marginal change in the catastrophe probability times the welfare gap between the two regimes. The framework is however not limited to these cases and applies to any distribution satisfying mild regularity conditions, nesting the standard social cost of carbon when shocks are exogenous.

Afterwards, I generalize the framework to a recursive infinite-horizon model, using the fact that the framework is an instance of a Markov decision process where tomorrow's transition probability depends not only on the current state but also on the current action. Although economists usually deal with regular Markov processes when working with recursive methods (see for example Stokey et al., 1989), there is also a literature extending these methods to Markov decision processes (Bertsekas, 1995; Hernández-Lerma & Lasserre, 1996). I then apply the framework to a modified DICE model, called the DICE-SE model, where there are climate shocks whose distribution shifts with emissions. In a calibrated model, the

stochastic externality raises the social cost of carbon by 6.2% in a conservative calibration where warming makes disasters more frequent but not more severe. If disasters are also allowed to be more severe, the social cost of carbon is instead increased by 8.6%.

This paper is not the first to model endogenous climate risk. A large economics literature already treats climate change as raising the probability of a discrete catastrophe, typically by modeling its arrival as a Poisson process whose hazard rate increases with pollution (Barro, 2015; Brausmann & Bretschger, 2024; Cai & Lontzek, 2019; Müller-Fürstenberger & Schumacher, 2015; van der Ploeg, 2014; van der Ploeg & de Zeeuw, 2018; Zemel, 2015). However, this literature is mostly concerned by a different kind of risk, namely the possibility of a catastrophic irreversible shift in climate dynamics, such as the collapse of major weather currents, the melting of major ice sheets, or the start of runaway climate feedbacks. This paper instead focuses on the change in the distribution of weather events that Earth sciences claim is already underway, such as floods, droughts, and wildfires. Rather than modeling climate change as an increase in the arrival probability of climate shocks, which are otherwise deterministic in their size, I consider a shift in the entire distribution of climate shocks. A parallel empirical literature is already beginning to estimate how climate change reshapes the distribution of weather shocks and their damages (Bakkensen & Barrage, 2026), so this paper connects this micro-level evidence to the macro-level question of optimal policy.

Section 2 introduces the three-period model, moving from standard externalities to stochastic externalities and closing with normal, Pareto, and regime-mixture examples. Section 3 generalizes to an infinite horizon and shows how to write the problem recursively. Section 4 describes the DICE-SE model, fits the climate shock distribution to 1980-2024 disaster losses, and quantifies the effect of the stochastic externality. Section 5 concludes.

2 Three-period model

2.1 Standard externalities

For simplicity, suppose that there are only three periods and economic activity generates a pollution stock S harming future productive capacity. I write the resource constraint of the economy in a general form allowing various specifications of climate-economy dynamics.

$$c_t + k_{t+1} = G(k_t, S_t, a_t, x_t)$$

$G(\cdot)$ is a differentiable resource function increasing in capital k_t and decreasing in the pollution stock S_t as well as the climate shock x_t . The vector a_t represents other controls, such as the abatement rate or another production input.

Capital, output, or another control affects emissions according to a differentiable function $e_t = \mathcal{E}(k_t, a_t)$ that contributes to the pollution stock. The pollution stock is a vector $S_t \in \mathbb{R}^n$ evolving according to a differentiable function $S_{t+1} = \mathcal{S}(S_t, e_t)$. Writing the pollution stock as a vector allows for multiple pollutants or a complex carbon cycle.

This is a simple yet general setup that represents a wide range of models in the literature. Although it does not consider consumption externalities, utility damages, or other features, these additions would be compatible with the model. As a concrete example, the DICE2013R model as written in Nordhaus and Sztorc (2013) maps onto this framework as follows¹.

$$\begin{aligned} G(k_t, S_t, a_t) &= \Delta [1 - \Omega(T_{\text{AT},t}) - \Lambda_t(a_t)] \tilde{Y}(k_t) + (1 - \delta)^\Delta k_t \\ \mathcal{E}(k_t, a_t) &= \sigma_t (1 - a_t) \tilde{Y}(k_t) + E_t^{\text{land}} \\ \mathcal{S}(S_t, e_t) &= \begin{pmatrix} b_{11}M_{\text{AT},t} + b_{21}M_{\text{U},t} + \zeta e_t \\ b_{12}M_{\text{AT},t} + b_{22}M_{\text{U},t} + b_{32}M_{\text{L},t} \\ b_{23}M_{\text{U},t} + b_{33}M_{\text{L},t} \\ T_{\text{AT},t} + c_1 [\eta \log_2(M_{\text{AT},t}/M_{\text{AT}}^{\text{pre}}) + F_t^{\text{EX}} - \lambda T_{\text{AT},t} - c_3 (T_{\text{AT},t} - T_{\text{OC},t})] \\ T_{\text{OC},t} + c_4 (T_{\text{AT},t} - T_{\text{OC},t}) \end{pmatrix} \end{aligned}$$

a_t is the emissions-control rate. $\mathcal{S}$ contains $S_t = (M_{\text{AT},t}, M_{\text{U},t}, M_{\text{L},t}, T_{\text{AT},t}, T_{\text{OC},t})$, where the first three are the carbon stock in three reservoirs and the last two are the temperatures of the atmosphere and the ocean, and emissions e_t in a system of equations. $\mathcal{E}$ is generated by post-abatement gross output and exogenous emissions and G is the output net of damages and abatement costs together with undepreciated capital. Section 4.1 modifies a form of this model to include stochastic externalities, but other IAMs could be used. Until then, I keep the model general for simplicity.

¹I use the 2013R version rather than more recent vintages for reader convenience as it is the latest version with an official user manual, though the structure remains the same across versions.

To fix ideas, consider the standard case without climate shocks. The planner solves:

$$\begin{aligned}
& \max_{\{c_t, k_{t+1}, a_t, S_{t+1}\}} u(c_0) + \beta u(c_1) + \beta^2 u(c_2) \\
& \text{s.t.} \\
& c_t + k_{t+1} = G(k_t, S_t, a_t) \quad t = 0, 1, 2 \\
& e_t = \mathcal{E}(k_t, a_t) \quad t = 0, 1 \\
& S_{t+1} = \mathcal{S}(S_t, e_t) \quad t = 0, 1
\end{aligned}$$

There is no fourth period, so there is no need to track emissions in $t = 2$. Attach $\beta^t u'(c_t)$ to the date- t resource constraint and μ_{t+1} to the stock transition. Since S_2 is terminal, its multiplier collects only the marginal damage the stock does to date-2 resources, and it is negative because G decreases in the stock.

$$\mu_2 = \beta^2 u'(c_2) \nabla_S G(k_2, S_2, a_2) \quad (1)$$

Capital, meanwhile, earns its marginal product and raises the stock through emissions.

$$u'(c_0) = \beta u'(c_1) G_k(k_1, S_1, a_1) + \mu_2^\top \mathcal{S}_e \mathcal{E}_k(k_1, a_1) \quad (2)$$

A unit of date-1 emissions moves the date-2 stock by $\mathcal{S}_e$, so contracting (1) with $\mathcal{S}_e$ turns the stock's shadow value into the marginal effect of that unit on date-2 resources, $\partial G_2 / \partial e_1 \equiv \nabla_S G(k_2, S_2, a_2)^\top \mathcal{S}_e$, giving $\mu_2^\top \mathcal{S}_e = \beta^2 u'(c_2) \partial G_2 / \partial e_1$. Substituting that into (2) and factoring out $\beta u'(c_1)$ gives the Euler condition between periods 0 and 1.

$$u'(c_0) = \beta u'(c_1) \left(G_k(k_1, S_1, a_1) + \underbrace{\beta \frac{u'(c_2)}{u'(c_1)} \frac{\partial G_2}{\partial e_1}}_{\text{Standard externality}} \mathcal{E}_k(k_1, a_1) \right)$$

The household Euler condition does not take externalities into account, so the second term in the parentheses is absent in its optimum. The planner would like to tax $\mathcal{E}$ at $\tau = -\beta \frac{u'(c_2)}{u'(c_1)} \frac{\partial G_2}{\partial e_1}$. This standard case also applies when resources depend on a climate shock x following an “exogenous” distribution, meaning one whose law does not respond to the pollution stock. The same three steps go through, with two changes. The resource constraints now hold shock by shock, so the multipliers become state contingent: μ_2 is chosen after x_1 is known and is therefore a function of it. And each multiplier is averaged over the shocks it is not yet measurable against. The condition for S_2 is the date-1 conditional expectation of its deterministic counterpart.

$$\mu_2(x_1) = \beta^2 \mathbb{E}_{x_2}[u'(c_2) \nabla_S G(k_2, S_2, a_2, x_2) \mid x_1] \quad (3)$$

The condition for k_1 is the date-0 expectation of its own, with the stock term carrying $\mu_2(x_1)$ inside the expectation because the shadow value of the stock now varies with the date-1 state.

$$u'(c_0) = \beta \mathbb{E}_{x_1}[u'(c_1) G_k(k_1, S_1, a_1, x_1)] + \mathbb{E}_{x_1}[\mu_2(x_1)^\top \mathcal{S}_e \mathcal{E}_k(k_1, a_1)] \quad (4)$$

Contracting (3) with $\mathcal{S}_e$ gives $\mu_2(x_1)^\top \mathcal{S}_e = \beta^2 \mathbb{E}_{x_2}[u'(c_2) \partial G_2 / \partial e_1 \mid x_1]$, the expected marginal damage of a date-1 unit of emissions conditional on the date-1 state. Substituting that into (4) and factoring $u'(c_1)$ out of the inner expectation gives the planner's Euler condition.

$$u'(c_0) = \beta \mathbb{E}_{x_1} \left[u'(c_1) \left(G_k(k_1, S_1, a_1, x_1) + \underbrace{\beta \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial G_2}{\partial e_1} \mid x_1 \right]}_{\text{Standard externality}} \mathcal{E}_k(k_1, a_1) \right) \right]$$

The optimal carbon tax decentralizing the planner's equilibrium is $\tau = -\beta \mathbb{E} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial G_2}{\partial e_1} \mid x_1 \right]$. Both taxes are Pigouvian, equal either to the discounted marginal damages of pollution or their expectation.

2.2 Stochastic externalities

Now consider the same model where the distribution of climate shocks shifts with pollution. I make the following assumptions about climate shocks.

Assumption 2.1 (Shock process).

Let $\mathcal{N} \subseteq \mathbb{R}^n$ be an open set containing the optimal pollution stocks. For each $S \in \mathcal{N}$, the date- t shock has distribution function $F(\cdot \mid x_{t-1}; S)$ on a possibly unbounded interval $[\underline{x}, \bar{x}]$ that does not vary with S , such that:

- i. F is continuous and strictly increasing in x on the support, with a density f strictly positive on its interior;
- ii. for each x , the maps $S \mapsto F(x \mid x_{t-1}; S)$ and $S \mapsto f(x \mid x_{t-1}; S)$ are continuously differentiable on $\mathcal{N}$, with gradients $\nabla_S F(x \mid x_{t-1}; S)$ and $\nabla_S f(x \mid x_{t-1}; S)$ in $\mathbb{R}^n$;
- iii. differentiation with respect to S and integration over x may be interchanged on the interior of the support.

Assumption 2.2 (Tail integrability).

At each infinite endpoint, $u(c_t(x)) \nabla_S F(x \mid x_{t-1}; S) \rightarrow 0$.

The planner solves the problem below, where I write the expectations in integral form to make explicit the dependence of the distribution on the pollution stock.

$$\begin{aligned} \max_{\{c_t, k_{t+1}, a_t, S_{t+1}\}} \quad & u(c_0) + \beta \int u(c_1(x_1)) dF(x_1 \mid x_0; S_1) \\ & + \beta^2 \iint u(c_2(x_1, x_2)) dF(x_2 \mid x_1; S_2) dF(x_1 \mid x_0; S_1) \end{aligned} \tag{5}$$

s.t.

$$\begin{aligned} c_0 + k_1 &= G(k_0, S_0, a_0, x_0) \\ c_1(x_1) + k_2(x_1) &= G(k_1, S_1, a_1, x_1) & \forall x_1 \\ c_2(x_1, x_2) &= G(k_2(x_1), S_2, a_2, x_2) & \forall x_1, x_2 \\ e_t &= \mathcal{E}(k_t, a_t) & t = 0, 1 \\ S_{t+1} &= \mathcal{S}(S_t, e_t) & t = 0, 1 \end{aligned}$$

Proposition 2.1 (Three-period carbon tax with stochastic externalities).

The planner can decentralize the solution to problem (5) with this tax in score form:

$$\tau(x_1) = \underbrace{-\beta \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial G_2}{\partial e_1} \middle| x_1 \right]}_{\text{Standard externality}} - \underbrace{\beta \text{Cov}_{x_2} \left(\frac{u(c_2)}{u'(c_1)}, \frac{\partial \ln f(x_2 | x_1; S_2)}{\partial e_1} \middle| x_1 \right)}_{\text{Stochastic externality}}$$

Which can also be written in quantile form:

$$\tau(x_1) = \underbrace{-\beta \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial G_2}{\partial e_1} \middle| x_1 \right]}_{\text{Standard externality}} + \underbrace{\beta \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} (-G_x(k_2, S_2, a_2, x_2)) \frac{\partial Q(p(x_2); S_2)}{\partial e_1} \middle| x_1 \right]}_{\text{Stochastic externality}}$$

Proof. See Appendix A. □

The setup nests the standard externality. When the distribution of climate shocks does not respond to the pollution stock, the second term disappears and the tax takes the standard Pigouvian form. The score is the proportional change in the likelihood of each shock realization, positive when a particular event is more likely and negative when it is less likely. Utility is weakly decreasing in climate shocks since a larger shock means lower output and therefore lower consumption. The covariance is negative when the two move in opposite directions on average, meaning emissions shift the probability of climate shocks towards worse outcomes. In this case, the corrective tax goes up.

The quantile form says the same thing about outcomes rather than probabilities. In its expectation, the first term is the stochastic discount factor, the second is the marginal damage of shocks on the resource function, and the third is the change in the shock quantile. Take the third quartile for example. This term asks whether the threshold below which three-quarters of shocks fall goes up or down after a change in emissions. If the threshold goes up on average, the tax goes up because the shock at every probability level is larger.

Depending on the distribution, the score form or the quantile form will be more convenient. Because some events might become more likely and others less likely, the sign of the correction is not obvious. Another way to write the stochastic part of the correction term is as follows.

$$\tau^{\text{stoch}}(x_1) = -\beta \int_{\underline{x}}^{\bar{x}} \frac{u'(c_2)}{u'(c_1)} (-G_x(k_2, S_2, a_2, x_2)) \frac{\partial F(x_2 | x_1; S_2)}{\partial e_1} dx_2 \quad (6)$$

The two following corollaries give conditions under which the correction is positive.

Corollary 2.1 (First-order stochastic dominance).

Let a unit of emissions worsen the distribution of climate shocks in the first-order stochastic dominance sense, so that $\partial F/\partial e_1 \leq 0$ everywhere. Equivalently, so that $\partial Q/\partial e_1 \geq 0$ at every probability level. Then, $\tau^{stoch}(x_1) \geq 0$. The opposite holds if the distribution improves.

Proof. $u'(c_2)(-G_x)/u'(c_1) \geq 0$ and $-\partial F/\partial e_1 \geq 0$, so the integrand in (6) is nonnegative. $\square$

Corollary 2.2 (Increase in risk).

Let $\mathcal{R}(x)$ be the integral of $\partial F/\partial e_1$ over $[\underline{x}, x]$ and let emissions induce a mean-preserving increase in risk so that $\mathcal{R}(x) \geq 0$ everywhere and $\mathcal{R}(\bar{x}) = 0$. If utility is concave, the resource function is concave in the shock and $\frac{u'(c_2)}{u'(c_1)}(-G_x)\mathcal{R} \rightarrow 0$ at the top of the support,

$$\tau^{stoch}(x_1) = \beta \int_{\underline{x}}^{\bar{x}} \left[-\frac{u''(c_2)}{u'(c_1)} G_x(k_2, S_2, a_2, x_2)^2 - \frac{u'(c_2)}{u'(c_1)} G_{xx}(k_2, S_2, a_2, x_2) \right] \mathcal{R}(x_2) dx_2 \geq 0$$

Proof. First, integrate (6) by parts.

$$\tau^{stoch} = -\beta \left[\frac{u'(c_2)}{u'(c_1)} (-G_x) \mathcal{R} \right]_{\underline{x}}^{\bar{x}} + \beta \int_{\underline{x}}^{\bar{x}} \frac{d}{dx_2} \left[\frac{u'(c_2)}{u'(c_1)} (-G_x) \right] \mathcal{R}(x_2) dx_2$$

Note that the first term disappears because $\mathcal{R}(\bar{x}) = 0$ by construction and I assumed $\frac{u'(c_2)}{u'(c_1)}(-G_x)\mathcal{R}$ is zero at the top. Differentiating the latter term and using $\partial c_2/\partial x_2 = G_x$ gives the bracket in the statement, which is nonnegative when $u'' \leq 0$ and $G_{xx} \leq 0$. $\square$

The first corollary can also be interpreted as saying that the tax rises when every quantile of the shock distribution has worse outcomes. The second corollary says that an increase in risk raises the tax when agents are risk averse or the resource function is concave in the shock. To build intuition, I present three illustrative examples. First, normal shocks, which are unlikely to represent the way actual climate shocks behave but are simple to understand. Second, Pareto shocks, highlighting the importance of tail behavior. Third, a two-regime mixture nesting the tipping-point approach.

As a first example, let $x \sim N(\mu(S), \sigma(S))$ and write $\mu_e \equiv \partial\mu/\partial e_1$ and $\sigma_e \equiv \partial\sigma/\partial e_1$ for the responses of the mean and the standard deviation to emissions. Take both to be positive, meaning emissions make shocks worse on average and more volatile. I derive the stochastic correction term in Appendix B. The correction splits into a mean-shift and a volatility channel, the latter further split into a risk-aversion and a damage-curvature channel.

$$\tau^{\text{stoch}}(x_1) = \underbrace{\beta \mu_e \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} (-G_x) \mid x_1 \right]}_{\text{Mean shift}} + \underbrace{\beta \sigma_e \sigma \mathbb{E}_{x_2} \left[-\frac{u''(c_2)}{u'(c_1)} G_x^2 - \frac{u'(c_2)}{u'(c_1)} G_{xx} \mid x_1 \right]}_{\text{Volatility}} \quad (7)$$

The mean-shift channel is always positive, while the volatility channel is positive when either utility is concave or the resource function is concave in the shock. Figure 1 illustrates these three channels. The mean-shift channel in panel (a) raises the carbon tax since climate shocks are now higher on average. The risk-aversion channel in panel (b) raises the tax when utility is concave. Although the normal distribution is symmetric and a variance increase therefore makes both low and high shocks more likely in equal measure, the utility loss in the right tail exceeds the gain in the left tail. If utility is linear, such that the agent is risk-neutral, this channel disappears because the agent weighs the extra gain and the extra loss equally. The damage-curvature channel in panel (c) depends on the curvature of the resource function in the shock. If resources are concave in the shock, this channel raises the tax since a wider distribution adds more damage in the right tail than it removes in the left.

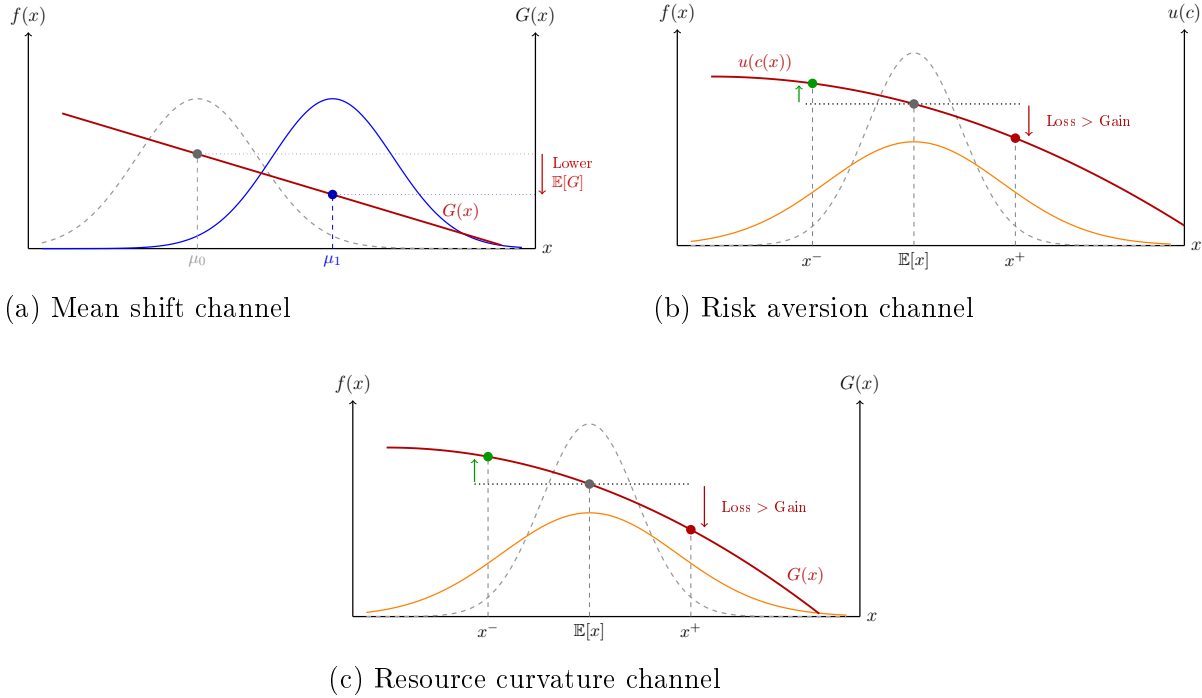


Figure 1: Stochastic externality channels for normally distributed shocks.

As a second example, let $x \sim \text{Pareto}(\alpha(S), x_m)$ with support $x_m > 0$ and write $\alpha_e \equiv \partial\alpha/\partial e_1$ for the response of the shape parameter to emissions. Take $\alpha_e < 0$ so that emissions make the tail heavier and $\alpha > 1$ so that the shock has a finite mean. I derive the stochastic correction term in Appendix C, which features a single tail-fattening channel.

$$\tau^{\text{stoch}}(x_1) = -\beta \frac{\alpha_e}{\alpha} \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} (-G_x) x_2 \ln \left(\frac{x_2}{x_m} \right) \middle| x_1 \right]$$

Figure 2 illustrates this channel. When the shape parameter α falls and the right tail grows heavier, each quantile of the shocks shifts right, as shown in panel (a). However, the shift is weighted by $w(x) \equiv x \ln(x/x_m)$, as shown in panel (b). This term grows without bound and places an increasing weight on rare catastrophic events. Depending on functional forms, the expectation may diverge and the stochastic externality term may become unbounded, echoing the "Dismal Theorem" (Weitzman, 2009).

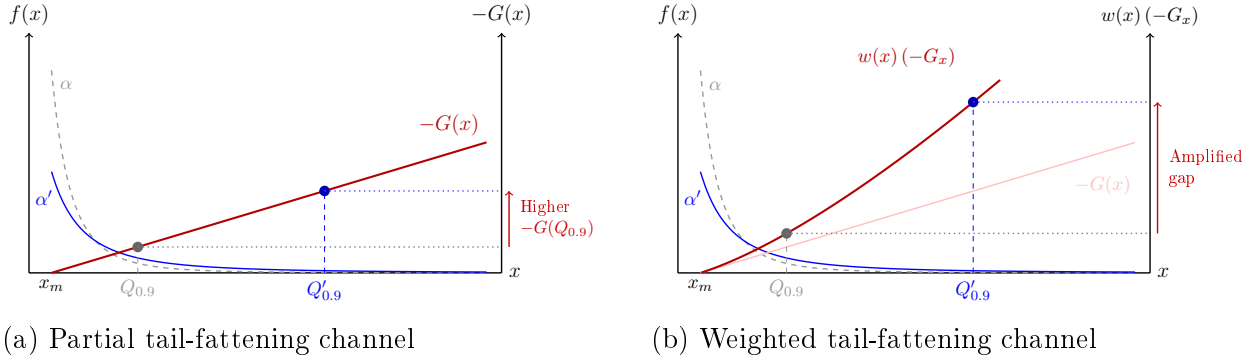


Figure 2: Pareto tail-fattening channel.

To see the implications of this weighting, consider $x \ln(x/x_m)$, ignoring the other terms in the expectation. If $\alpha > 1$, then $\mathbb{E}[x] = \frac{\alpha x_m}{\alpha - 1}$. For the expectation of $x \ln(x/x_m)$, note that $y \equiv \ln(x/x_m)$ follows an exponential distribution with rate α .

$$\mathbb{E} \left[x \ln \left(\frac{x}{x_m} \right) \right] = x_m \mathbb{E}[y e^y]$$

$\mathbb{E}[y e^y]$ is the derivative of the exponential distribution's moment generating function with respect to its argument s , evaluated at $s = 1$, which is admissible because $\alpha > 1$.

$$M_y = \frac{\alpha}{\alpha - s} \implies \mathbb{E}[y e^y] = \frac{\partial M_y}{\partial s} \bigg|_{s=1} = \frac{\alpha}{(\alpha - 1)^2}$$

Hence the weighted moment is the mean inflated by a factor that depends only on the tail.

$$\mathbb{E} \left[x \ln \left(\frac{x}{x_m} \right) \right] = \frac{\alpha x_m}{(\alpha - 1)^2} = \frac{\mathbb{E}[x]}{\alpha - 1}$$

Both moments are finite when $\alpha > 1$, but the weighted moment grows faster as $\alpha \rightarrow 1$. Therefore, the stochastic externality term may come to dominate as the tail fattens.

As a third example, let the shock be drawn from a normal regime with distribution F_0 with probability $1 - \pi(S)$ and a catastrophic regime with distribution F_1 with probability $\pi(S)$. Both regimes satisfy Assumption 2.1.

$$F(x_2 | x_1; S_2) = (1 - \pi(S_2)) F_0(x_2) + \pi(S_2) F_1(x_2) \quad (8)$$

Write $\pi_e \equiv \partial\pi/\partial e_1$ for the effect of emissions on catastrophe probability and suppose $\pi_e > 0$. Let the catastrophic regime be worse in the first-order stochastic dominance sense, $F_1 \leq F_0$. I derive the following correction term in Appendix D.

$$\tau^{\text{stoch}}(x_1) = \beta \pi_e \frac{\mathbb{E}_0[u(c_2)] - \mathbb{E}_1[u(c_2)]}{u'(c_1)} \quad (9)$$

This is the marginal change in the catastrophe probability times the welfare gap between the two regimes, similar to the form the tipping point literature often works with, where the carbon price is a hazard rate times the value lost on arrival (Polasky et al., 2011; van der Ploeg, 2014; van der Ploeg & de Zeeuw, 2018). The difference here is that this literature uses a degenerate version of (8), in which F_0 and F_1 are point masses and damage is deterministic given the regime, which violates item (i) of Assumption 2.1² The general result however still holds as we only need $F_1 - F_0$ to disappear at both ends and $u(c_2(x))$ to be continuous.

In the next section, I generalize the three-period framework to an arbitrary number of periods and show that the problem can be written in a recursive form.

²This paper's results can be generalized to allow for degenerate distributions by weakening items (i) and (ii) of Assumption 2.1, replacing (i) with the requirement that $\nabla_S F$ be of bounded variation in x and keeping only the F half of (ii). The derivations would then use Riemann-Stieltjes integrals. However, this would lead to less interpretable results, not allowing for the conceptually convenient score and quantile forms.

3 Multiple period model

Extend the three-period model to an infinite horizon and denote by $x^t \equiv (x_0, \dots, x_t)$ the history of climate shocks up to t . The sequential problem can be written as follows.

$$\max_{\{c_t(x^t), k_{t+1}(x^t), a_t(x^t)\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \beta^t \int \cdots \int u(c_t(x^t)) dF(x_t | x_{t-1}; S_t) \cdots dF(x_0 | x_{-1}; S_0) \right\}$$

s.t.

$$\begin{aligned} c_t(x^t) + k_{t+1}(x^t) &= G(k_t(x^{t-1}), S_t(x^{t-1}), a_t(x^t), x_t) & \forall t \\ e_t &= \mathcal{E}(k_t(x^{t-1}), a_t(x^t)) & \forall t \\ S_{t+1}(x^t) &= \mathcal{S}(S_t(x^{t-1}), e_t) & \forall t \\ x_{t+1} &\sim F(\cdot | x_t; S_{t+1}(x^t)) & \forall t \end{aligned}$$

This problem is unwieldy as written, but restating it in recursive form makes it more akin to the simple three-period problem. Although standard recursive methods used in economics typically assume exogenous shocks following a Markov process (e.g., Stokey et al., 1989), recursivity still holds here. The shocks follow a Markov decision process, where tomorrow's transition probability depends not only on the current state but also on the current action (e.g., Bertsekas, 1995; Hernández-Lerma & Lasserre, 1996). In the notation of Hernández-Lerma and Lasserre (1996, Definition 2.2.1), a Markov decision process is described by $(X, A, \{A(x)\}, Q, r)$, where X is the state space, A is the action space, $A(x)$ is the set of feasible actions given the state, Q is the transition law, and r is the reward³. Here, the state is (k_t, S_t, x_t) , the action is (c_t, k_{t+1}, a_t) , $A(x)$ is determined by the resource constraint, the transition probability is $F(\cdot | x_t; S_{t+1}(x^t))$, and r is utility.

Hernández-Lerma and Lasserre (1996, Chap. 4) show when an infinite-horizon problem has a recursive form. By their Assumption 4.2.1, the reward must be upper semicontinuous, bounded above, and sup-compact⁴, and the transition probability must be weakly continuous⁵. If utility is continuous, then upper semicontinuity is immediately satisfied. If the feasible set is compact-valued, then sup-compactness is also satisfied. Boundedness is more restrictive, but is satisfied when either utility is bounded above or the resource function is bounded above along feasible allocations. By their Assumption 4.2.2, there must be a policy function such that the value function is finite for every state. By their Proposition 4.3.1, this is satisfied when either (i) the reward is bounded, (ii) the reward is dominated by some function of the state that on average grows less than the discount rate, or (iii) the discounted sum of

³The definition is stated with a cost rather than a reward in the textbook, but the authors note that the two can be used. Indeed, in example 1.3.2, they use utility as a reward.

⁴In the textbook, the problem is stated as a cost minimization, so the conditions are flipped.

⁵The textbook requires strong continuity, but notes that the assumption is copied from variant b2 of Condition 3.3.4, which also allows for weak continuity in its variant b1.

rewards is finite even under the worst feasible actions. If their Assumptions 4.2.1 and 4.2.2 are satisfied, the sequential problem is solved by the following Bellman equation.

$$V(k, S, x) = \max_{k', a} \left\{ u(G(k, S, a, x) - k') + \beta \int V(k', S', x') dF(x' | x; S') \right\} \quad (10)$$

s.t.

$$S' = \mathcal{S}(S, \mathcal{E}(k, a))$$

Note that shock endogeneity has not added any new state variables compared to the standard stochastic problem because the pollution stock, which now affects the shock distribution, already affected output in the previous problem. From the perspective of today's agent, since the current pollution stock was already decided last period, the current shock distribution might as well be exogenous. Hence, the infinite-horizon problem is not conceptually different from the three-period problem and every result from Section 2 carries over.

Proposition 3.1 (Infinite-horizon carbon tax with stochastic externalities).

Let Assumption 2.1 and Assumptions 4.2.1-4.2.2 of Hernández-Lerma and Lasserre (1996) hold. The planner can decentralize the solution to problem (10) with this tax in score form:

$$\tau_t = - \underbrace{\sum_{j=0}^{\infty} \beta^{j+1} \mathbb{E}_t \left[\frac{u'(c_{t+j+1})}{u'(c_t)} \frac{\partial G_{t+j+1}}{\partial e_t} \right]}_{\text{Standard externality}} - \underbrace{\sum_{j=0}^{\infty} \beta^{j+1} \text{Cov}_t \left(\frac{V_{t+j+1}}{u'(c_t)}, \frac{\partial \ln f(x_{t+j+1} | x_{t+j}; S_{t+j+1})}{\partial e_t} \right)}_{\text{Stochastic externality}}$$

Which can also be written in quantile form:

$$\tau_t = - \underbrace{\sum_{j=0}^{\infty} \beta^{j+1} \mathbb{E}_t \left[\frac{u'(c_{t+j+1})}{u'(c_t)} \frac{\partial G_{t+j+1}}{\partial e_t} \right]}_{\text{Standard externality}} + \underbrace{\sum_{j=0}^{\infty} \beta^{j+1} \mathbb{E}_t \left[\frac{u'(c_{t+j+1})}{u'(c_t)} (-G_x) \frac{\partial Q(p(x_{t+j+1}); S_{t+j+1})}{\partial e_t} \right]}_{\text{Stochastic externality}}$$

Proof. See Appendix E. □

The interpretation of the formulas is the same as in the three-period case, though the score form is now a covariance between the continuation value and the score rather than between future utility and the score. That is, the score form looks at how the probability of shocks changes while the quantile form looks at how the shocks at each quantile change. The two are equivalent, but one may be more convenient than the other depending on the distribution. As in the three-period case, the stochastic term can be written as follows.

$$\tau_t^{\text{stoch}} = - \sum_{j=0}^{\infty} \beta^{j+1} \mathbb{E}_t \left[\int_{\underline{x}}^{\bar{x}} \frac{u'(c_{t+j+1})}{u'(c_t)} (-G_x) \frac{\partial F(x | x_{t+j}; S_{t+j+1})}{\partial e_t} dx \right]$$

From here, Corollaries 2.1 and 2.2 carry over from the three-period case to give the conditions where the correction term is unambiguously positive.

Corollary 3.1 (First-order stochastic dominance at every period).

Let a unit of the date- t flow worsen the distribution of climate shocks in the first-order stochastic dominance sense at every period, so that $\partial F(x | x_{t+j}; S_{t+j+1})/\partial e_t \leq 0$ everywhere for every j . Equivalently, so that $\partial Q(p; S_{t+j+1})/\partial e_t \geq 0$ at every probability level and every horizon. Then, $\tau_t^{\text{stoch}} \geq 0$. The opposite holds if the distribution improves at every horizon.

Proof. $u'(c_{t+j+1}) (-G_x) / u'(c_t) \geq 0$ and $-\partial F/\partial e_t \geq 0$, so integrands are nonnegative. $\square$

Corollary 3.2 (Increase in risk at every period).

Let $\mathcal{R}_{t+j+1}(x)$ be the integral of $\partial F(\cdot | x_{t+j}; S_{t+j+1})/\partial e_t$ over $[\underline{x}, x]$ and let a unit of the date- t flow induce a mean-preserving increase in risk at every period, so that $\mathcal{R}_{t+j+1}(x) \geq 0$ everywhere and $\mathcal{R}_{t+j+1}(\bar{x}) = 0$ for every j . If the value function is concave in the shock and $\frac{u'(c_{t+j+1})}{u'(c_t)} (-G_x) \mathcal{R}_{t+j+1} \rightarrow 0$ at the top of the support at every period,

$$\tau_t^{\text{stoch}} = \sum_{j=0}^{\infty} \beta^{j+1} \mathbb{E}_t \left[\int_{\underline{x}}^{\bar{x}} \left(-\frac{1}{u'(c_t)} \frac{\partial^2 V(k_{t+j+1}, S_{t+j+1}, x)}{\partial x^2} \right) \mathcal{R}_{t+j+1}(x) dx \right] \geq 0$$

Proof. Integrate by parts the integral in the sum, as in Corollary 2.2.

$$\tau_t^{\text{stoch}} = \sum_{j=0}^{\infty} \beta^{j+1} \mathbb{E}_t \left[- \left[\frac{u'(c_{t+j+1})}{u'(c_t)} (-G_x) \mathcal{R}_{t+j+1} \right]_{\underline{x}}^{\bar{x}} + \int_{\underline{x}}^{\bar{x}} \frac{d}{dx} \left[\frac{u'(c_{t+j+1})}{u'(c_t)} (-G_x) \right] \mathcal{R}_{t+j+1}(x) dx \right]$$

The first term disappears because $\mathcal{R}_{t+j+1}(\underline{x}) = 0$ by construction and I assumed the weight is zero against $\mathcal{R}_{t+j+1}$ at the top. By the envelope condition,

$$\frac{u'(c_{t+j+1})}{u'(c_t)} (-G_x) = -\frac{1}{u'(c_t)} \frac{\partial V(k_{t+j+1}, S_{t+j+1}, x)}{\partial x}$$

Differentiating it in the shock gives the bracket in the statement, which is nonnegative when V is concave in the shock. A sum of nonnegative terms is nonnegative. $\square$

In infinite horizon, the second corollary folds the risk-aversion and damage-curvature channel into the curvature of the value function in the shock, $\partial^2 V / \partial x^2$. Otherwise, the two corollaries are essentially the same as in the three-period model.

Before showing the implication of stochastic externalities on the social cost of carbon in a calibrated integrated assessment model, I extend the model to allow for recursive preferences (Epstein & Zin, 1989), which allows disentangling risk aversion and intertemporal substitution. I replace the objective in (10) by a recursive aggregator.

$$V(k, S, x) = \max_{k', a} \{u_\psi(G(k, S, a, x) - k') + \beta \mathcal{C}(V(k', S', x'))\}$$

$$\mathcal{C} \equiv \left(\int V(k', S', x')^\vartheta dF(x' | x; S') \right)^{1/\vartheta}$$

Here $u_\psi(c) = c^{1-1/\psi} / (1 - 1/\psi)$, where ψ is the elasticity of intertemporal substitution, γ the coefficient of relative risk aversion, and $\vartheta \equiv (1 - \gamma) / (1 - 1/\psi)$. The certainty equivalent $\mathcal{C}$ replaces the conditional expectation of (10), which is the special case $\vartheta = 1$. Define:

$$\widehat{\mathbb{E}}_t[Z_{t+1}] \equiv \mathbb{E}_t \left[\left(\frac{V_{t+1}}{\mathcal{C}_t} \right)^{\vartheta-1} Z_{t+1} \right] \quad \widehat{\mathbb{E}}_t^{(j)}[Z_{t+j}] \equiv \mathbb{E}_t \left[\prod_{i=1}^j \left(\frac{V_{t+i}}{\mathcal{C}_{t+i-1}} \right)^{\vartheta-1} Z_{t+j} \right]$$

This risk-adjusted expectation overweights states with low continuation value when $\gamma > 1/\psi$. Aside from this change, the optimal carbon tax formulas remain essentially the same.

Proposition 3.2 (Infinite-horizon carbon tax under recursive preferences).

Let Assumption 2.1 and Assumptions 4.2.1-4.2.2 of Hernández-Lerma and Lasserre (1996) hold. The planner can decentralize the solution to problem (10) under Epstein-Zin preferences with this tax in score form, which reduces to Proposition 3.1 when $\vartheta = 1$:

$$\tau_t = - \sum_{j=0}^{\infty} \beta^{j+1} \widehat{\mathbb{E}}_t^{(j+1)} \left[\frac{u'_\psi(c_{t+j+1})}{u'_\psi(c_t)} \frac{\partial G_{t+j+1}}{\partial e_t} \right] - \sum_{j=0}^{\infty} \frac{\beta^{j+1}}{\vartheta} \widehat{\mathbb{E}}_t^{(j)} \left[\mathcal{C}_{t+j}^{1-\vartheta} \text{Cov}_{t+j} \left(\frac{V_{t+j+1}^\vartheta}{u'_\psi(c_t)}, \frac{\partial \ln f_{t+j+1}}{\partial e_t} \right) \right]$$

Which can also be written in quantile form:

$$\tau_t = - \sum_{j=0}^{\infty} \beta^{j+1} \widehat{\mathbb{E}}_t^{(j+1)} \left[\frac{u'_\psi(c_{t+j+1})}{u'_\psi(c_t)} \frac{\partial G_{t+j+1}}{\partial e_t} \right] + \sum_{j=0}^{\infty} \beta^{j+1} \widehat{\mathbb{E}}_t^{(j+1)} \left[\frac{u'_\psi(c_{t+j+1})}{u'_\psi(c_t)} (-G_x) \frac{\partial Q(p(x_{t+j+1}); S_{t+j+1})}{\partial e_t} \right]$$

Proof. See Appendix F. □

4 Quantitative results

4.1 DICE-SE model and calibration

To illustrate the effect of stochastic externalities on the social cost of carbon, I take the benchmark DICE model and modify it to include stochastic climate losses whose distribution depends on the temperature. I call the model DICE-SE, standing for stochastic externalities. More specifically, I take the recursive DICE model as written in Cai and Lontzek (2019) and modify the damage function such that net output is as follows.

$$Y_t = \frac{(1 - \Lambda_t) \tilde{Y}_t}{1 + a_1 T_{AT,t} + a_2 T_{AT,t}^2 + x_t - m(T_{AT,t})},$$

Here, $x_t \geq 0$ is the year's aggregate disaster loss, measured in the data as a share of gross world output. $m(T) \equiv \mathbb{E}[x \mid T]$ is its conditional mean, so that the shock term $x_t - m(T_{AT,t})$ averages zero at every temperature. The idea is that the DICE damage function accurately captures the average effect of climate change but not its variation. I call this mean-preserving specification Config A and use it as the baseline. Config B instead sets $m \equiv 0$, treating disaster losses as absent from the standard DICE calibration.

I calibrate the distribution of climate shocks $x \sim F(x; \theta(T))$ on 1980-2024 global disaster losses from EM-DAT, expressed as shares of same-year gross world output (Pielke et al., 2008), and use the NOAA billion-dollar disaster database for the United States as an alternative calibration (Smith, 2020). The shock follows a lognormal body below a fixed threshold $\bar{u}$, B , and a generalized Pareto tail above it, G . The choice of having a generalized Pareto distribution at the tail is motivated by extreme value theory (Coles, 2001; Embrechts et al., 1997). The lognormal is truncated at the threshold and rescaled so that the distribution function remains continuous throughout. The exceedance probability $\pi(T)$, the probability that a year's loss exceeds the threshold, depends on temperature.

$$F(x; \theta(T)) = \begin{cases} (1 - \pi(T)) \frac{B(x; \mu(T), \sigma_B(T))}{B(\bar{u}; \mu(T), \sigma_B(T))} & x \leq \bar{u}, \\ (1 - \pi(T)) + \pi(T) G(x - \bar{u}; \varsigma(T), \xi) & x > \bar{u}, \end{cases}$$

Table 1 reports how each parameter depends on temperature, together with the fitted estimates. Each link is centered at 1°C, the mean temperature anomaly over the sample, while T_a is the mean anomaly in the years in which the loss exceeds the threshold. The threshold $\bar{u}$ is set at the 80th percentile of annual losses and λ_0 is set so that the exceedance probability is 0.20 at T_a . The body and tail parameters are estimated by maximum likelihood while and the frequency slope b is estimated by a Poisson regression of the annual number of large events on temperature.

Table 1: Dependence of the shock distribution’s parameters on temperature. T is the global mean surface temperature anomaly relative to 1880-1919, in °C.

	Link to temperature	Estimate
<i>Body</i> (lognormal, $x \leq \bar{u}$)		
Mean of $\ln x$	$\mu(T) = \mu_0 + \mu_1 \cdot (T - 1)$	$\mu_0 = -6.52, \mu_1 = 0.68$
Log-standard-deviation	$\ln \sigma_B(T) = s_0 + s_1 \cdot (T - 1)$	$s_0 = -0.86, s_1 = -0.10$
<i>Splice point</i>		
Threshold	$\bar{u}$ constant in T	0.192% of gross world output
<i>Tail</i> (generalized Pareto, $x > \bar{u}$)		
Exceedance probability	$\pi(T) = 1 - e^{-\lambda(T)}$	
Arrival rate of tail years	$\lambda(T) = \lambda_0 e^{b \cdot (T - T_a)}$	$\lambda_0 = 0.223, b = 0.59, T_a = 1.04$
Log-scale	$\ln \varsigma(T) = c + d \cdot (T - 1)$	$c = -7.62, d = 0$
Shape	ξ constant in T	$\xi = 0.68$

The fitted tail index is $\xi = 0.68$, with a bootstrap interval of $[0.54, 0.82]$. This tail is heavy enough that the loss share has an infinite variance, though a finite mean. Fitting the tail to the NOAA data instead while pooling all US hazards gives $\xi = 1.05$, where the mean is no longer finite. Using US tropical cyclones alone gives $\xi = 0.82$. The aggregate loss data do not cleanly identify how the tail scale responds to warming, so I set $d = 0$ as a conservative baseline and consider $d = 0.28$ and $d = 0.53$ as two alternatives. These two raise the tail scale by 32% and 70% per degree of warming.

This spliced family effectively nests the examples of Section 2. It is a two-regime mixture as in (8), with the lognormal body as the normal regime and the generalized Pareto tail as the catastrophic regime. Warming raises the regime weight $\pi(T)$, shifts the body up as in the mean-shift channel of (7), and widens the tail regime when $d > 0$. I solve DICE-SE by dynamic programming over 600 annual periods. The social cost of carbon is separated in a Pigouvian term and a stochastic-externality premium.

4.2 Results

The social cost of carbon in 2005 is \$36.50/tCO₂, of which \$34.37 is the Pigouvian term and \$2.13 is the stochastic-externality premium. The stochastic externality therefore grows the price of carbon by 6.2%. In 2050, the premium rises to \$2.50 and becomes \$3.82 in 2100.

When the tail index is made to be more sensitive to warming, the premium grows. In the bootstrap interval for the calibrated value, it runs from \$190 at $\xi = 0.54$ to \$2.51 at $\xi = 0.82$. At the NOAA value of $\xi = 1.05$ where the loss share has no finite mean, it reaches \$3.54.

At tail scale $d = 0.28$ and $d = 0.53$, the premium is \$2.91 and \$3.93, or 8.0% and 10.7% of the tax. The premium here grows much faster than at the baseline, to \$9.84 and \$21.26 by 2100, or 5.8% and 12.6% of the tax

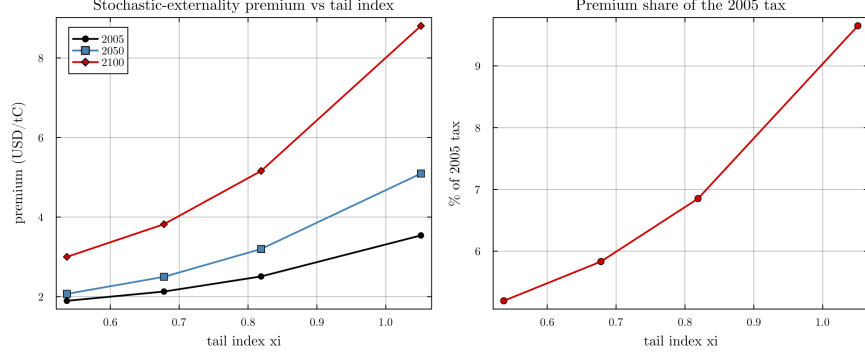


Figure 3: The stochastic-externality premium as a function of ξ . Left: the premium in 2005, 2050 and 2100. Right: the premium as a percent of the 2005 tax.

Table 2: 2005 decomposition and selected path values across the calibration grid

Run	Tax 2005	Pigou 2005	Premium 2005	Share 2005	Premium 2050	Premium 2100	T_{AT} 2100
Baseline	36.50	34.37	2.13	5.8	2.50	3.82	2.48
$d = 0.28$ (central)	36.58	33.67	2.91	8.0	4.71	9.84	2.48
$d = 0.53$ (high)	36.78	32.85	3.93	10.7	8.24	21.26	2.49
$\xi = 0.537$ (CI lower)	36.48	34.59	1.90	5.2	2.07	3.00	2.48
$\xi = 0.819$ (CI upper)	36.61	34.10	2.51	6.9	3.20	5.16	2.48
$\xi = 1.051$ (NOAA)	36.68	33.14	3.54	9.6	5.09	8.81	2.48
Persistence $\rho = 0.146$	36.34	34.24	2.10	5.8	2.45	3.74	2.48

Epstein-Zin preferences separate risk aversion and elasticity of intertemporal substitution (Epstein & Zin, 1989). Table 3 reports the premium on a grid over γ and ψ , which is computed by solving the model with and without stochastic externalities and then taking the difference. Risk aversion has little effect on the premium whereas the elasticity of intertemporal substitution has a moderate effect, due to the conservative baseline calibration.

Table 3: Epstein-Zin stochastic-externality premium

γ	ψ	Tax 2005	Pigou 2005	Premium 2005	Share 2005	Premium 2100
2	0.5	39.88	37.76	2.12	5.31	3.64
2	1.5	98.62	94.44	4.18	4.24	9.77
10	0.5	39.89	37.76	2.12	5.32	3.67
10	1.5	98.63	94.44	4.18	4.24	9.79
10	2.0	116.57	111.74	4.82	4.14	11.99
15	1.5	98.63	94.44	4.19	4.25	9.80

5 Conclusion

In this paper, I develop a theoretical framework for analyzing the effect of stochastic externalities on the social cost of carbon. Stochastic externalities are defined as the effect of pollution on the distribution of future climate shocks, reflecting the "distributional shift" perspective of climate risk where climate variables such as CO₂ concentration or average temperature make extreme events not only more frequent but also more severe. This perspective is common in the Earth sciences and is beginning to be empirically assessed in economics, yet its implications for the social cost of carbon have not yet been formally characterized. This paper fills this gap.

In a three-period model later extended to an infinite-horizon, I show that the social cost of carbon decomposes into a standard Pigouvian term and a novel stochastic externality correction term. The latter can be expressed in a number of convenient forms, such as a score form representing how the probability of future climate shocks changes with pollution or an equivalent quantile form representing how climate shocks at each quantiles become better or worse with pollution. I provide three examples to illustrate the channels, with a normal distribution, a Pareto distribution, and a regime-mixture distribution.

To show the quantitative implications of stochastic externalities, I extend the benchmark DICE model into the DICE-SE model, modifying its damage function to introduce stochastic climate shocks whose distribution depends on the temperature. The results suggest that stochastic externalities generate a small to moderate premium in the social cost of carbon, depending on the calibration of the climate shock distribution. Currently, the results are merely illustrative and future work will implement a more rigorous calibration methodology.

There is a number of important extensions for future work, such implementing the framework in a heterogeneous-agent model. It would be interesting to analyze optimal mitigation with adaptation or insurance where, as per Schumacher (2019), the former is viewed as a public good and the latter is viewed as a private good. In this analysis, I could evaluate the tradeoffs between *ex ante* mitigation ("prevention is better than cure") and *ex post* adaptation or insurance in the context of climate change, where correcting for stochastic externalities is an indirect way to correct for lack of adaptation or insurance.

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Model documentation for DICE-2013R. Detailed equations are in Section III: emissions eq. (10) p. 14; carbon cycle eqs. (13)-(15) p. 16; radiative forcing eq. (16) p. 17; two-layer temperature eqs. (17)-(18) p. 17; production and damages eqs. (4)-(6) pp. 10-12; Hotelling constraint eq. (11) p. 14. Known errata: eq. (9) p. 13 prints $K(t) =$

- $I(t) - \delta_K K(t-1)$, omitting the $(1-\delta_K)$ factor; eq. (5) p. 10 prints ψ_1 twice where the second should be ψ_2 . The damage and abatement nesting in eq. (4) p. 10 differs from the GAMS source, which computes $Y_{\text{gross}}*(1 - D - \lambda)$. No equivalent manual exists for DICE-2016R; that vintage is documented only in NBER Working Paper 22933 and Nordhaus (2017, PNAS). Title page states only the copyright line and the website dicemodel.net; "Yale University" is the conventional attribution, not a publisher statement in the document.
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A Proof of Proposition 2.1

First, I solve the household's problem. The household is atomistic, taking the pollution stock and the distribution of shocks as given. It faces a tax τ_t on its emissions, whose revenue is returned as a lump-sum transfer T_t . The household's problem is as follows.

$$\begin{aligned}
& \max_{\{c_0, c_1, c_2, k_1, k_2, a_0, a_1, a_2\}} u(c_0) + \beta \int u(c_1(x_1)) dF(x_1 | x_0; S_1) \\
& \quad + \beta^2 \iint u(c_2(x_1, x_2)) dF(x_2 | x_1; S_2) dF(x_1 | x_0; S_1) \\
& \quad \text{s.t.} \\
& \quad c_0 + k_1 = G(k_0, S_0, a_0, x_0) - \tau_0 e_0 + T_0 \\
& \quad c_1(x_1) + k_2(x_1) = G(k_1, S_1, a_1, x_1) - \tau_1 e_1 + T_1 \quad \forall x_1 \\
& \quad c_2(x_1, x_2) = G(k_2(x_1), S_2, a_2, x_2) + T_2 \quad \forall x_1, x_2 \\
& \quad e_t = \mathcal{E}(k_t, a_t) \quad t = 0, 1
\end{aligned}$$

Substituting emissions by their expression, the Lagrangian is:

$$\begin{aligned}
\mathcal{L} = & u(c_0) + \beta \int u(c_1(x_1)) dF(x_1 | x_0; S_1) + \beta^2 \iint u(c_2(x_1, x_2)) dF(x_2 | x_1; S_2) dF(x_1 | x_0; S_1) \\
& + \lambda_0 [G(k_0, S_0, a_0, x_0) - \tau_0 \mathcal{E}(k_0, a_0) + T_0 - c_0 - k_1] \\
& + \beta \int \lambda_1(x_1) [G(k_1, S_1, a_1, x_1) - \tau_1 \mathcal{E}(k_1, a_1) + T_1 - c_1(x_1) - k_2(x_1)] dF(x_1 | x_0; S_1) \\
& + \beta^2 \iint \lambda_2(x_1, x_2) [G(k_2(x_1), S_2, a_2, x_2) + T_2 - c_2(x_1, x_2)] dF(x_2 | x_1; S_2) dF(x_1 | x_0; S_1)
\end{aligned}$$

The first-order conditions are:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial c_0} : \lambda_0 &= u'(c_0) \\
\frac{\partial \mathcal{L}}{\partial c_1(x_1)} : \lambda_1(x_1) &= u'(c_1(x_1)) \quad \forall x_1 \\
\frac{\partial \mathcal{L}}{\partial c_2(x_1, x_2)} : \lambda_2(x_1, x_2) &= u'(c_2(x_1, x_2)) \quad \forall x_1, x_2 \\
\frac{\partial \mathcal{L}}{\partial k_1} : \lambda_0 &= \beta \int \lambda_1(x_1) (G_k(k_1, S_1, a_1, x_1) - \tau_1 \mathcal{E}_k(k_1, a_1)) f(x_1 | x_0; S_1) dx_1 \\
\frac{\partial \mathcal{L}}{\partial k_2(x_1)} : \lambda_1(x_1) &= \beta \int \lambda_2(x_1, x_2) G_k(k_2(x_1), S_2, a_2, x_2) f(x_2 | x_1; S_2) dx_2 \quad \forall x_1
\end{aligned}$$

$$\begin{aligned}
\nabla_{a_0} \mathcal{L} : \nabla_a G(k_0, S_0, a_0, x_0) &= \tau_0 \nabla_a \mathcal{E}(k_0, a_0) \\
\nabla_{a_1(x_1)} \mathcal{L} : \nabla_a G(k_1, S_1, a_1, x_1) &= \tau_1 \nabla_a \mathcal{E}(k_1, a_1) & \forall x_1 \\
\nabla_{a_2(x_1, x_2)} \mathcal{L} : \nabla_a G(k_2(x_1), S_2, a_2, x_2) &= \mathbf{0} & \forall x_1, x_2
\end{aligned}$$

Since a is a vector, the last three first-order conditions are expressed as vectors. From here, the household's Euler condition between periods 0 and 1 is:

$$u'(c_0) = \beta \int u'(c_1) (G_k(k_1, S_1, a_1, x_1) - \tau_1 \mathcal{E}_k(k_1, a_1)) f(x_1 | x_0; S_1) dx_1$$

Next, let us move on to the planner's problem, whose Lagrangian is:

$$\begin{aligned}
\mathcal{L} = & u(c_0) + \beta \int u(c_1(x_1)) dF(x_1 | x_0; S_1) + \beta^2 \iint u(c_2(x_1, x_2)) dF(x_2 | x_1; S_2) dF(x_1 | x_0; S_1) \\
& + \lambda_0 [G(k_0, S_0, a_0, x_0) - c_0 - k_1] \\
& + \beta \int \lambda_1(x_1) [G(k_1, S_1, a_1(x_1), x_1) - c_1(x_1) - k_2(x_1)] dF(x_1 | x_0; S_1) \\
& + \beta^2 \iint \lambda_2(x_1, x_2) [G(k_2(x_1), S_2, a_2(x_1, x_2), x_2) - c_2(x_1, x_2)] dF(x_2 | x_1; S_2) dF(x_1 | x_0; S_1) \\
& + \mu_1^\top [\mathcal{S}(S_0, \mathcal{E}(k_0, a_0)) - S_1] + \int \mu_2(x_1)^\top [\mathcal{S}(S_1, \mathcal{E}(k_1, a_1(x_1))) - S_2(x_1)] dF(x_1 | x_0; S_1)
\end{aligned}$$

The first-order conditions are:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial c_0} : \lambda_0 &= u'(c_0) \\
\frac{\partial \mathcal{L}}{\partial c_1(x_1)} : \lambda_1(x_1) &= u'(c_1(x_1)) & \forall x_1 \\
\frac{\partial \mathcal{L}}{\partial c_2(x_1, x_2)} : \lambda_2(x_1, x_2) &= u'(c_2(x_1, x_2)) & \forall x_1, x_2
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial k_1} : \lambda_0 &= \beta \int \lambda_1(x_1) G_k(k_1, S_1, a_1(x_1), x_1) f(x_1 | x_0; S_1) dx_1 \\
& + \int \mu_2(x_1)^\top \mathcal{S}_e \mathcal{E}_k(k_1, a_1(x_1)) f(x_1 | x_0; S_1) dx_1 \\
\frac{\partial \mathcal{L}}{\partial k_2(x_1)} : \lambda_1(x_1) &= \beta \int \lambda_2(x_1, x_2) G_k(k_2(x_1), S_2, a_2(x_1, x_2), x_2) f(x_2 | x_1; S_2) dx_2 & \forall x_1
\end{aligned}$$

$$\begin{aligned}
\nabla_{a_0} \mathcal{L} : \lambda_0 \nabla_a G(k_0, S_0, a_0, x_0) + \mu_1^\top \mathcal{S}_e \nabla_a \mathcal{E}(k_0, a_0) &= \mathbf{0} \\
\nabla_{a_1(x_1)} \mathcal{L} : \beta \lambda_1(x_1) \nabla_a G(k_1, S_1, a_1(x_1), x_1) + \mu_2(x_1)^\top \mathcal{S}_e \nabla_a \mathcal{E}(k_1, a_1(x_1)) &= \mathbf{0} \quad \forall x_1 \\
\nabla_{a_2(x_1, x_2)} \mathcal{L} : \nabla_a G(k_2(x_1), S_2, a_2(x_1, x_2), x_2) &= \mathbf{0} \quad \forall x_1, x_2
\end{aligned}$$

$$\begin{aligned}
\nabla_{S_1} \mathcal{L} : \mu_1 &= \beta \int \lambda_1(x_1) \nabla_S G(k_1, S_1, a_1(x_1), x_1) f(x_1 \mid x_0; S_1) dx_1 \\
&+ \int \mathcal{S}_S^\top \mu_2(x_1) f(x_1 \mid x_0; S_1) dx_1 \\
&+ \int \left(\beta u(c_1) + \beta^2 \int u(c_2) f(x_2 \mid x_1; S_2) dx_2 \right) \nabla_{S_1} f(x_1 \mid x_0; S_1) dx_1 \\
\nabla_{S_2(x_1)} \mathcal{L} : \mu_2(x_1) &= \beta^2 \int \lambda_2(x_1, x_2) \nabla_S G(k_2(x_1), S_2, a_2(x_1, x_2), x_2) f(x_2 \mid x_1; S_2) dx_2 \\
&+ \beta^2 \int u(c_2) \nabla_{S_2} f(x_2 \mid x_1; S_2) dx_2 \quad \forall x_1
\end{aligned}$$

In the last two first-order conditions, the pollution stock affects the objective twice, one time through the resource function and another time through the climate shock density. Substituting λ_t and factoring $u'(c_1)$ out of the second integral in the condition for k_1 ,

$$u'(c_0) = \beta \mathbb{E}_{x_1} \left[u'(c_1) \left(G_k(k_1, S_1, a_1, x_1) + \frac{\mu_2(x_1)^\top \mathcal{S}_e}{\beta u'(c_1)} \mathcal{E}_k(k_1, a_1) \right) \right] \quad (11)$$

Taking the transpose of both sides of the condition for S_2 and multiplying them by $\frac{\mathcal{S}_e}{\beta u'(c_1)}$.

$$\frac{\mu_2(x_1)^\top \mathcal{S}_e}{\beta u'(c_1)} = \beta \int \frac{u'(c_2)}{u'(c_1)} \nabla_S G(k_2, S_2, a_2, x_2)^\top \mathcal{S}_e f(x_2 \mid x_1; S_2) dx_2 + \beta \int \frac{u(c_2)}{u'(c_1)} \nabla_{S_2} f(x_2 \mid x_1; S_2)^\top \mathcal{S}_e dx_2$$

By the multivariate chain rule, $\frac{\partial G_2}{\partial e_1} = \nabla_S G(k_2, S_2, a_2, x_2)^\top \mathcal{S}_e$ and $\frac{\partial f}{\partial e_1} = \nabla_{S_2} f(x_2 \mid x_1; S_2)^\top \mathcal{S}_e$.

$$\frac{\mu_2(x_1)^\top \mathcal{S}_e}{\beta u'(c_1)} = \beta \int \frac{u'(c_2)}{u'(c_1)} \frac{\partial G_2}{\partial e_1} f(x_2 \mid x_1; S_2) dx_2 + \beta \int \frac{u(c_2)}{u'(c_1)} \frac{\partial f(x_2 \mid x_1; S_2)}{\partial e_1} dx_2 \quad (12)$$

So that both terms read as expectations under the same density, I multiply and divide the second term by f using $\partial \ln f / \partial e_1 = (\partial f / \partial e_1) / f$.

$$\frac{\mu_2(x_1)^\top \mathcal{S}_e}{\beta u'(c_1)} = \beta \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial G_2}{\partial e_1} \mid x_1 \right] + \beta \mathbb{E}_{x_2} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln f(x_2 \mid x_1; S_2)}{\partial e_1} \mid x_1 \right]$$

Then, I substitute in (11).

$$u'(c_0) = \beta \mathbb{E}_{x_1} \left[u'(c_1) \left(G_k(k_1, S_1, a_1, x_1) + \beta \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial G_2}{\partial e_1} + \frac{u(c_2)}{u'(c_1)} \frac{\partial \ln f}{\partial e_1} \middle| x_1 \right] \mathcal{E}_k(k_1, a_1) \right) \right]$$

Let us recall again the household's Euler condition, written in expectation form.

$$u'(c_0) = \beta \mathbb{E}_{x_1} [u'(c_1) (G_k(k_1, S_1, a_1, x_1) - \tau_1 \mathcal{E}_k(k_1, a_1))]$$

The planner's and household's Euler conditions coincide when the tax is set as follows.

$$\tau(x_1) = -\beta \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial G_2}{\partial e_1} \middle| x_1 \right] - \beta \mathbb{E}_{x_2} \left[\frac{u(c_2)}{u'(c_1)} \frac{\partial \ln f}{\partial e_1} \middle| x_1 \right] \quad (13)$$

I can write this tax in two convenient forms. First, a score form. By part (iii) of Assumption 2.1, I can switch the differential and the integral in the stochastic term so that $\int \nabla_{S_2} f dx_2 = \nabla_{S_2} \int f dx_2 = 0$. Therefore, the stochastic term can be expressed as a covariance between utility and the score of the shock distribution.⁶

$$\tau(x_1) = \underbrace{-\beta \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial G_2}{\partial e_1} \middle| x_1 \right]}_{\text{Standard externality}} - \underbrace{\beta \text{Cov}_{x_2} \left(\frac{u(c_2)}{u'(c_1)}, \frac{\partial (\ln f(x_2 | x_1; S_2))}{\partial e_1} \middle| x_1 \right)}_{\text{Stochastic externality}}$$

Second, it can be expressed in quantile form. Integrate the stochastic term in (12) by parts.

$$\int_{\underline{x}}^{\bar{x}} u(c_2) \frac{\partial f}{\partial e_1} dx_2 = \left[u(c_2) \frac{\partial F}{\partial e_1} \right]_{\underline{x}}^{\bar{x}} - \int_{\underline{x}}^{\bar{x}} u'(c_2) G_x(k_2, S_2, a_2, x_2) \frac{\partial F}{\partial e_1} dx_2 \quad (14)$$

The bracket term disappears because the cdf is always 0 at the lower bound and 1 at the upper bound, such that its derivative is zero at both ends. Next, let $Q(p; S_2) \equiv F^{-1}(p | x_1; S_2)$ be the quantile function. Differentiating $F(Q(p; S_2) | x_1; S_2) = p$ at fixed p gives $\frac{\partial F}{\partial e_1} = -f \frac{\partial Q}{\partial e_1}$. Substituting into (14) and then multiplying by $-\beta/u'(c_1)$ as in (13),

$$\tau(x_1) = \underbrace{-\beta \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} \frac{\partial G_2}{\partial e_1} \middle| x_1 \right]}_{\text{Standard externality}} + \underbrace{\beta \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} (-G_x(k_2, S_2, a_2, x_2)) \frac{\partial Q(p(x_2); S_2)}{\partial e_1} \middle| x_1 \right]}_{\text{Stochastic externality}}$$

⁶ $\text{Cov}(x, y) = \mathbb{E}[xy] - \mathbb{E}[x]\mathbb{E}[y]$ and the score has mean zero, so $\text{Cov}(x, y) = \mathbb{E}[xy]$.

B Derivation of the normal decomposition

Write $\mathcal{U}(x_2) \equiv u(c_2)/u'(c_1)$ for welfare in date-1 consumption units and $z \equiv (x_2 - \mu)/\sigma$ for the standardized shock. The normal log-density is $\ln f = -\ln \sigma - \frac{1}{2} \ln 2\pi - \frac{1}{2} z^2$, so that $\partial \ln f / \partial \mu = z/\sigma$ and $\partial \ln f / \partial \sigma = (z^2 - 1)/\sigma$. Holding x_2 fixed and differentiating,

$$\frac{\partial \ln f}{\partial e_1} = \frac{\mu_e}{\sigma} z + \frac{\sigma_e}{\sigma} (z^2 - 1) \quad (15)$$

Substituting (15) into the stochastic term of (13),

$$\tau^{\text{stoch}}(x_1) = -\beta \frac{\mu_e}{\sigma} \mathbb{E}_{x_2}[\mathcal{U}z \mid x_1] - \beta \frac{\sigma_e}{\sigma} \mathbb{E}_{x_2}[\mathcal{U}(z^2 - 1) \mid x_1] \quad (16)$$

Both expectations follow from Stein's lemma. For $z \sim N(0, 1)$ and any differentiable g whose relevant expectations are finite, $\mathbb{E}[g(z)z] = \mathbb{E}[g'(z)]$. Applying this to g' in place of g gives $\mathbb{E}[g(z)(z^2 - 1)] = \mathbb{E}[g''(z)]$. Take $g(z) = \mathcal{U}(\mu + \sigma z)$, so that $g'(z) = \sigma \mathcal{U}_x$ and $g''(z) = \sigma^2 \mathcal{U}_{xx}$. The resource constraint gives $\partial c_2 / \partial x_2 = G_x$, so the two derivatives of $\mathcal{U}$ are:

$$\mathcal{U}_x = \frac{u'(c_2)}{u'(c_1)} G_x \quad \mathcal{U}_{xx} = \frac{u''(c_2)}{u'(c_1)} G_x^2 + \frac{u'(c_2)}{u'(c_1)} G_{xx} \quad (17)$$

Substituting (17) into (16), the factors of σ cancel and both signs flip.

$$\tau^{\text{stoch}}(x_1) = \beta \mu_e \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} (-G_x) \mid x_1 \right] + \beta \sigma_e \mathbb{E}_{x_2} \left[-\frac{u''(c_2)}{u'(c_1)} G_x^2 - \frac{u'(c_2)}{u'(c_1)} G_{xx} \mid x_1 \right]$$

The two terms in the volatility expectation are the risk-aversion and resource-curvature channels of Corollary 2.2.

C Derivation of the Pareto decomposition

The Pareto distribution function is $F(x; \alpha) = 1 - (x_m/x)^\alpha$ on $x \geq x_m$, with quantile function $Q(p; \alpha) = x_m (1-p)^{-1/\alpha}$. If I take the logarithm of the quantile function and then differentiate it with respect to the shape parameter α at a fixed probability level p ,

$$\frac{\partial Q}{\partial \alpha} = Q \frac{\ln(1-p)}{\alpha^2} \quad (18)$$

At $x_2 = Q(p; \alpha)$, the definition of the quantile function gives $\ln(1-p) = -\alpha \ln(x_2/x_m)$. Substituting that into (18) and applying the chain rule through α ,

$$\frac{\partial Q}{\partial e_1} = \alpha_e \frac{\partial Q}{\partial \alpha} = -\frac{\alpha_e}{\alpha} x_2 \ln\left(\frac{x_2}{x_m}\right) \quad (19)$$

The stochastic term of the quantile form is

$$\beta \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} (-G_x) \frac{\partial Q}{\partial e_1} \mid x_1 \right]$$

Substituting (19) and taking the constant $-\alpha_e/\alpha$ outside the expectation gives

$$\tau^{\text{stoch}}(x_1) = -\beta \frac{\alpha_e}{\alpha} \mathbb{E}_{x_2} \left[\frac{u'(c_2)}{u'(c_1)} (-G_x) x_2 \ln\left(\frac{x_2}{x_m}\right) \mid x_1 \right]$$

D Derivation of the regime-mixture decomposition

Only the probability responds to emissions, so differentiating gives the shift in the distribution.

$$\frac{\partial F}{\partial e_1} = \pi_e (F_1(x_2) - F_0(x_2)) \quad (20)$$

Substituting (20) into the shift form (6),

$$\tau^{\text{stoch}}(x_1) = -\beta \pi_e \int_{\underline{x}}^{\bar{x}} \frac{u'(c_2)}{u'(c_1)} (-G_x) (F_1 - F_0) dx_2 \quad (21)$$

The resource constraint gives $\partial c_2 / \partial x_2 = G_x$, so the marginal welfare cost of a shock is the derivative of welfare in date-1 consumption units with its sign reversed.

$$\frac{u'(c_2)}{u'(c_1)} (-G_x) = -\frac{d}{dx_2} \left[\frac{u(c_2)}{u'(c_1)} \right] \quad (22)$$

Substituting (22) into (21) and integrating by parts, the boundary term vanishes because F_0 and F_1 are distribution functions on the same support, so $F_1 - F_0$ is zero at both ends.

$$\int_{\underline{x}}^{\bar{x}} \frac{u'(c_2)}{u'(c_1)} (-G_x) (F_1 - F_0) dx_2 = \int_{\underline{x}}^{\bar{x}} \frac{u(c_2)}{u'(c_1)} (f_1 - f_0) dx_2 = \frac{\mathbb{E}_1[u(c_2)] - \mathbb{E}_0[u(c_2)]}{u'(c_1)} \quad (23)$$

Substituting (23) into (21) gives (9). Since $F_1 \leq F_0$, the catastrophic regime is worse in the first-order stochastic dominance sense and $\mathbb{E}_1[u(c_2)] < \mathbb{E}_0[u(c_2)]$. With $\pi_e > 0$, the correction is positive.

E Proof of Proposition 3.1

I go straight to the planner's problem, whose Lagrangian is as follows.

$$\mathcal{L} = u(c) + \beta \int V(k', S', x') dF(x' | x; S') + \lambda [G(k, S, a, x) - c - k'] + (\mu')^\top [\mathcal{S}(S, \mathcal{E}(k, a)) - S']$$

Writing $\mathcal{S}_S \equiv \nabla_S \mathcal{S}(S, e)$ and $\mathcal{S}_e \equiv \partial \mathcal{S}(S, e) / \partial e$ for the Jacobians of the transition,

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c} : \lambda &= u'(c) \\ \frac{\partial \mathcal{L}}{\partial k'} : \lambda &= \beta \int \frac{\partial V(k', S', x')}{\partial k'} dF(x' | x; S') \\ \nabla_a \mathcal{L} : \lambda \nabla_a G(k, S, a, x) + (\mu')^\top \mathcal{S}_e \nabla_a \mathcal{E}(k, a) &= \mathbf{0} \\ \nabla_{S'} \mathcal{L} : \mu' &= \beta \int \nabla_{S'} V(k', S', x') dF(x' | x; S') + \beta \int V(k', S', x') \nabla_{S'} f(x' | x; S') dx' \end{aligned}$$

The envelope conditions are:

$$\begin{aligned} \frac{\partial V}{\partial k} &= \lambda G_k(k, S, a, x) + (\mu')^\top \mathcal{S}_e \mathcal{E}_k(k, a) \\ \nabla_S V &= \lambda \nabla_S G(k, S, a, x) + \mathcal{S}_S^\top \mu' \\ \frac{\partial V}{\partial x} &= \lambda G_x(k, S, a, x) + \beta \int V(k', S', x') \frac{\partial f(x' | x; S')}{\partial x} dx' \end{aligned}$$

Substituting $\lambda = u'(c)$ into the capital envelope condition and then into the condition for k' gives the planner's Euler condition:

$$u'(c_t) = \beta \mathbb{E}_t \left[u'(c_{t+1}) \left(G_k(k_{t+1}, S_{t+1}, a_{t+1}, x_{t+1}) + \frac{\mu_{t+2}^\top \mathcal{S}_{e,t+1}}{u'(c_{t+1})} \mathcal{E}_k(k_{t+1}, a_{t+1}) \right) \right] \quad (24)$$

Substituting $\nabla_{S'} V$ in the first-order condition for S' using the envelope condition,

$$\mu_{t+1} = \beta \mathbb{E}_t [u'(c_{t+1}) \nabla_S G_{t+1} + \mathcal{S}_{S,t+1}^\top \mu_{t+2}] + \beta \int V_{t+1} \nabla_{S_{t+1}} f_{t+1} dx_{t+1}$$

Taking the transpose of both sides and multiplying them by $\mathcal{S}_{e,t}$,

$$\mu_{t+1}^\top \mathcal{S}_{e,t} = \beta \mathbb{E}_t [u'(c_{t+1}) \nabla_S G_{t+1}^\top \mathcal{S}_{e,t} + \mu_{t+2}^\top \mathcal{S}_{S,t+1} \mathcal{S}_{e,t}] + \beta \int V_{t+1} \nabla_{S_{t+1}} f_{t+1}^\top \mathcal{S}_{e,t} dx_{t+1}$$

By the multivariate chain rule, $\frac{\partial G_{t+1}}{\partial e_t} = \nabla_S G_{t+1}^\top \mathcal{S}_{e,t}$ and $\frac{\partial f_{t+1}}{\partial e_t} = \nabla_{S_{t+1}} f_{t+1}^\top \mathcal{S}_{e,t}$.

$$\mu_{t+1}^\top \mathcal{S}_{e,t} = \beta \mathbb{E}_t \left[u'(c_{t+1}) \frac{\partial G_{t+1}}{\partial e_t} + \mu_{t+2}^\top \frac{\partial S_{t+2}}{\partial e_t} \right] + \beta \int V_{t+1} \frac{\partial f_{t+1}}{\partial e_t} dx_{t+1}$$

From here, I can iteratively substitute $\mu_{t+j+1}^\top \partial S_{t+j+1} / \partial e_t$ and divide by $u'(c_t)$.

$$\frac{\mu_{t+1}^\top \mathcal{S}_{e,t}}{u'(c_t)} = \sum_{j=0}^{\infty} \beta^{j+1} \mathbb{E}_t \left[\frac{u'(c_{t+j+1})}{u'(c_t)} \frac{\partial G_{t+j+1}}{\partial e_t} + \int \frac{V_{t+j+1}}{u'(c_t)} \frac{\partial f_{t+j+1}}{\partial e_t} dx_{t+j+1} \right]$$

The inner integral of the second term is not an expectation like the outer integral because it does not integrate against a density. I can however turn it into an expectation by multiplying and dividing by f , using $\partial \ln f_{t+j+1} / \partial e_t = (\partial f_{t+j+1} / \partial e_t) / f_{t+j+1}$. By the law of iterated expectations, the inner integral folds into the outer integral.

$$\frac{\mu_{t+1}^\top \mathcal{S}_{e,t}}{u'(c_t)} = \sum_{j=0}^{\infty} \beta^{j+1} \mathbb{E}_t \left[\frac{u'(c_{t+j+1})}{u'(c_t)} \frac{\partial G_{t+j+1}}{\partial e_t} + \frac{V_{t+j+1}}{u'(c_t)} \frac{\partial \ln f_{t+j+1}}{\partial e_t} \right]$$

I substitute $\frac{\mu_{t+1}^\top \mathcal{S}_{e,t}}{u'(c_t)}$ in (24).

$$\begin{aligned} u'(c_t) = \beta \mathbb{E}_t & \left[u'(c_{t+1}) \left(G_k(k_{t+1}, S_{t+1}, a_{t+1}, x_{t+1}) \right. \right. \\ & + \sum_{j=0}^{\infty} \beta^{j+1} \mathbb{E}_{t+1} \left[\frac{u'(c_{t+j+2})}{u'(c_{t+1})} \frac{\partial G_{t+j+2}}{\partial e_{t+1}} \right] \mathcal{E}_k(k_{t+1}, a_{t+1}) \\ & \left. \left. + \sum_{j=0}^{\infty} \beta^{j+1} \mathbb{E}_{t+1} \left[\frac{V_{t+j+2}}{u'(c_{t+1})} \frac{\partial \ln f_{t+j+2}}{\partial e_{t+1}} \right] \mathcal{E}_k(k_{t+1}, a_{t+1}) \right) \right] \end{aligned}$$

The planner's and household's Euler conditions coincide when the tax is set as follows.

$$\tau_t = - \sum_{j=0}^{\infty} \beta^{j+1} \mathbb{E}_t \left[\frac{u'(c_{t+j+1})}{u'(c_t)} \frac{\partial G_{t+j+1}}{\partial e_t} \right] - \sum_{j=0}^{\infty} \beta^{j+1} \mathbb{E}_t \left[\frac{V_{t+j+1}}{u'(c_t)} \frac{\partial \ln f_{t+j+1}}{\partial e_t} \right]$$

As in the three-period case, I can write this in score form. By part (iii) of Assumption 2.1, I can switch the differential and the integral in the stochastic term so that the stochastic term at each horizon can be expressed as a covariance between the value function and the score of the shock distribution.

$$\tau_t = \underbrace{-\sum_{j=0}^{\infty} \beta^{j+1} \mathbb{E}_t \left[\frac{u'(c_{t+j+1})}{u'(c_t)} \frac{\partial G_{t+j+1}}{\partial e_t} \right]}_{\text{Standard externality}} - \underbrace{\sum_{j=0}^{\infty} \beta^{j+1} \text{Cov}_t \left(\frac{V_{t+j+1}}{u'(c_t)}, \frac{\partial \ln f_{t+j+1}}{\partial e_t} \right)}_{\text{Stochastic externality}}$$

Likewise, it can be expressed in quantile form. Integrate by parts the stochastic term, using the envelope condition for the shock.

$$\int_{\underline{x}}^{\bar{x}} V_{t+j+1} \frac{\partial f_{t+j+1}}{\partial e_t} dx_{t+j+1} = \left[V_{t+j+1} \frac{\partial F_{t+j+1}}{\partial e_t} \right]_{\underline{x}}^{\bar{x}} - \int_{\underline{x}}^{\bar{x}} u'(c_{t+j+1}) G_x \frac{\partial F_{t+j+1}}{\partial e_t} dx_{t+j+1}$$

The bracket term disappears because the cdf is always 0 at the lower bound and 1 at the upper bound. Let $Q(p; S_{t+j+1}) \equiv F^{-1}(p \mid x_{t+j}; S_{t+j+1})$ be the quantile function. Differentiating $F(Q(p; S_{t+j+1}) \mid x_{t+j}; S_{t+j+1}) = p$ at fixed p gives $\partial F_{t+j+1} / \partial e_t = -f_{t+j+1} \partial Q / \partial e_t$. Hence,

$$\tau_t = \underbrace{-\sum_{j=0}^{\infty} \beta^{j+1} \mathbb{E}_t \left[\frac{u'(c_{t+j+1})}{u'(c_t)} \frac{\partial G_{t+j+1}}{\partial e_t} \right]}_{\text{Standard externality}} + \underbrace{\sum_{j=0}^{\infty} \beta^{j+1} \mathbb{E}_t \left[\frac{u'(c_{t+j+1})}{u'(c_t)} (-G_x) \frac{\partial Q(p(x_{t+j+1}); S_{t+j+1})}{\partial e_t} \right]}_{\text{Stochastic externality}}$$

F Proof of Proposition 3.2

The derivation is Appendix E with the conditional expectation replaced by the certainty equivalent, so the Lagrangian carries u_ψ in place of u and $\beta \mathcal{C}$ in place of $\beta \int V dF$.

$$\mathcal{L} = u_\psi(c) + \beta \mathcal{C} + \lambda [G(k, S, a, x) - c - k'] + (\mu')^\top [\mathcal{S}(S, \mathcal{E}(k, a)) - S']$$

Only the conditions that differentiate through $\mathcal{C}$ change. Differentiate the certainty equivalent in the form $\mathcal{C}^\vartheta = \int V^\vartheta dF$.

$$\begin{aligned} \frac{\partial \mathcal{C}}{\partial k'} &= \widehat{\mathbb{E}} \left[\frac{\partial V(k', S', x')}{\partial k'} \right] \\ \nabla_{S'} \mathcal{C} &= \widehat{\mathbb{E}}[\nabla_{S'} V(k', S', x')] + \frac{1}{\vartheta} \mathcal{C}^{1-\vartheta} \int V(k', S', x')^\vartheta \nabla_{S'} f dx' \end{aligned}$$

The first-order conditions are:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c} : \lambda &= u'_\psi(c) \\ \frac{\partial \mathcal{L}}{\partial k'} : \lambda &= \beta \widehat{\mathbb{E}} \left[\frac{\partial V(k', S', x')}{\partial k'} \right] \\ \nabla_a \mathcal{L} : \lambda \nabla_a G(k, S, a, x) + (\mu')^\top \mathcal{S}_e \nabla_a \mathcal{E}(k, a) &= \mathbf{0} \\ \nabla_{S'} \mathcal{L} : \mu' &= \beta \widehat{\mathbb{E}}[\nabla_{S'} V(k', S', x')] + \beta \frac{1}{\vartheta} \mathcal{C}^{1-\vartheta} \int V(k', S', x')^\vartheta \nabla_{S'} f dx' \end{aligned}$$

The envelope conditions are unchanged.

$$\begin{aligned} \frac{\partial V}{\partial k} &= \lambda G_k(k, S, a, x) + (\mu')^\top \mathcal{S}_e \mathcal{E}_k(k, a) \\ \nabla_S V &= \lambda \nabla_S G(k, S, a, x) + \mathcal{S}_S^\top \mu' \\ \frac{\partial V}{\partial x} &= \lambda G_x(k, S, a, x) \end{aligned}$$

From here, I can derive the planner's Euler condition.

$$u'_\psi(c_t) = \beta \widehat{\mathbb{E}}_t \left[u'_\psi(c_{t+1}) \left(G_k(k_{t+1}, S_{t+1}, a_{t+1}, x_{t+1}) + \frac{\mu_{t+2}^\top \mathcal{S}_{e,t+1}}{u'_\psi(c_{t+1})} \mathcal{E}_k(k_{t+1}, a_{t+1}) \right) \right]$$

Substitute $\nabla_{S'} V$ in the condition for S' using the envelope condition, take the transpose, multiply by $\mathcal{S}_{e,t}$ and apply the chain rule as in Appendix E. Multiplying and dividing by f_{t+1} turns the integral into a covariance with the score.

$$\mu_{t+1}^\top \mathcal{S}_{e,t} = \beta \widehat{\mathbb{E}}_t \left[u'_\psi(c_{t+1}) \frac{\partial G_{t+1}}{\partial e_t} + \mu_{t+2}^\top \frac{\partial S_{t+2}}{\partial e_t} \right] + \frac{\beta}{\vartheta} \mathcal{C}_t^{1-\vartheta} \text{Cov}_t \left(V_{t+1}^\vartheta, \frac{\partial \ln f_{t+1}}{\partial e_t} \right)$$

Iterating forward and substituting in the Euler condition,

$$\tau_t = - \sum_{j=0}^{\infty} \beta^{j+1} \widehat{\mathbb{E}}_t^{(j+1)} \left[\frac{u'_\psi(c_{t+j+1})}{u'_\psi(c_t)} \frac{\partial G_{t+j+1}}{\partial e_t} \right] - \sum_{j=0}^{\infty} \frac{\beta^{j+1}}{\vartheta} \widehat{\mathbb{E}}_t^{(j)} \left[\mathcal{C}_{t+j}^{1-\vartheta} \text{Cov}_{t+j} \left(\frac{V_{t+j+1}^\vartheta}{u'_\psi(c_t)}, \frac{\partial \ln f_{t+j+1}}{\partial e_t} \right) \right]$$

For the quantile form, write the covariance as an integral and integrate by parts. The envelope condition gives $\partial(V_{t+j+1}^\vartheta)/\partial x = \vartheta V_{t+j+1}^{\vartheta-1} u'_\psi(c_{t+j+1}) G_x$.

$$\begin{aligned} \frac{1}{\vartheta} \mathcal{C}_{t+j}^{1-\vartheta} \int_{\underline{x}}^{\bar{x}} V_{t+j+1}^\vartheta \frac{\partial f_{t+j+1}}{\partial e_t} dx_{t+j+1} &= \frac{1}{\vartheta} \mathcal{C}_{t+j}^{1-\vartheta} \left[V_{t+j+1}^\vartheta \frac{\partial F_{t+j+1}}{\partial e_t} \right]_{\underline{x}}^{\bar{x}} \\ &\quad - \mathcal{C}_{t+j}^{1-\vartheta} \int_{\underline{x}}^{\bar{x}} V_{t+j+1}^{\vartheta-1} u'_\psi(c_{t+j+1}) G_x \frac{\partial F_{t+j+1}}{\partial e_t} dx_{t+j+1} \end{aligned}$$

The bracket term disappears since the cdf is 0 at the lower bound and 1 at the upper bound. Substituting $\partial F_{t+j+1}/\partial e_t = -f_{t+j+1} \partial Q/\partial e_t$,

$$\frac{1}{\vartheta} \mathcal{C}_{t+j}^{1-\vartheta} \text{Cov}_{t+j} \left(V_{t+j+1}^\vartheta, \frac{\partial \ln f_{t+j+1}}{\partial e_t} \right) = - \widehat{\mathbb{E}}_{t+j} \left[u'_\psi(c_{t+j+1}) (-G_x) \frac{\partial Q(p(x_{t+j+1}); S_{t+j+1})}{\partial e_t} \right]$$

From here,

$$\tau_t = - \sum_{j=0}^{\infty} \beta^{j+1} \widehat{\mathbb{E}}_t^{(j+1)} \left[\frac{u'_\psi(c_{t+j+1})}{u'_\psi(c_t)} \frac{\partial G_{t+j+1}}{\partial e_t} \right] + \sum_{j=0}^{\infty} \beta^{j+1} \widehat{\mathbb{E}}_t^{(j+1)} \left[\frac{u'_\psi(c_{t+j+1})}{u'_\psi(c_t)} (-G_x) \frac{\partial Q(p(x_{t+j+1}); S_{t+j+1})}{\partial e_t} \right]$$

At $\vartheta = 1$, $\widehat{\mathbb{E}} = \mathbb{E}$ and I get Proposition 3.1.